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The promise and challenge of solar volumetric absorbers

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Abstract

This study investigates the potential performance of volumetric absorbers as a function of geometric and material properties, aiming to identify the best absorber design parameters and the highest efficiency that may be expected. A simplified one-dimensional model was used to represent a planar slab of ceramic foam absorber, with local thermal non-equilibrium and effective volumetric properties. Three approaches for modeling the radiative transfer were considered, and the S_4 discrete ordinates model was selected based on validation against a detailed Monte-Carlo simulation. The boundary conditions were investigated in detail. This model is simple enough for fast computation and parametric study, yet reasonably realistic to represent real absorbers. The results reveal several guidelines to improve the absorber performance. Optimization of geometry (porosity and characteristic pore diameter) is insufficient to reach high efficiency. A significant increase in convection heat transfer is required, beyond the normal behavior of ceramic foams. A reduction in the thermal conductivity of the absorber material is also needed to maintain the desired temperature distribution. Finally, spectral selectivity of the absorber material can also help to further increase the absorber efficiency, in contrast to the common opinion that it is effective only at low temperatures. With a combination of these measures, absorber efficiencies may be increased for example from around 70% to 90% for air heating to 1000 °C under incident flux of 800 kW/m^2.

Keywords: Solar receiver; Volumetric effect; Porous medium

1. Introduction

Electricity generation from solar energy by thermo-mechanical conversion is currently limited in worldwide implementation. A major reason is the relatively low conversion efficiency (typically 15–18% annual average), which contributes to the high cost of the produced electricity, 2–3 times higher than electricity produced from conventional fossil fuels (Pitz-Paal et al., 2012). This level of performance and cost is achieved today in solar thermal power plant technologies (parabolic trough and power tower) that are based on steam cycles at moderate temperatures of 400–550 °C. The solar dish-Stirling technology is capable of achieving much higher efficiency, but it has not been able so far to demonstrate the reliability and cost that would make it a serious contender, and it is considered a niche solution for small distributed generation plants. A breakthrough in the competitiveness of utility-scale solar thermal electricity may occur if the conversion efficiency from sunlight to electricity can be significantly increased. An important candidate to achieve high conversion efficiency is solar heating of air for high-temperature Combined Cycles (gas and steam turbines operating in series).

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Nomenclature

C	empirical constant in Eq. (4)
C_p	specific heat (J kg^{-1} K^{-1})
d	typical pore size (m), $d = 1/PPC$
d_m	typical passage size, $d_m = \sqrt{4\phi/\pi}/PPC$
e_b	blackbody emissive power (W m^{-2})
F	blackbody fraction function
$\overline{F}$	complimentary blackbody fraction, $\overline{F}(\lambda) = 1 - F(\lambda)$
G	incident radiation (W m^{-2})
h_v	volumetric convection coefficient (W m^{-3} K^{-1})
i	radiation intensity (W m^{-2} sr^{-1})
k	thermal conductivity (W m^{-1} K^{-1})
L	absorber thickness (m)
$\dot{m}$	mass flow rate per unit area (kg s^{-1} m^{-2})
Nu_v	volumetric Nusselt number ($=h_{vre}d^2 k_f^{-1}$)
p	pressure (Pa)
PPC	pores per cm
q	heat flux (W m^{-2})
q_R	radiative heat flux (W m^{-2})
Re	Reynolds number ($=\dot{m}\, d\mu^{-1}$)
T	temperature (K)
u	velocity (m s^{-1})
w	quadrature weight
w'	summed quadrature weight

Greek

α	surface absorptance and emittance (–)
α_v	volumetric absorption coefficient (m^{-1})
β	extinction coefficient (m^{-1})
λ_c	cutoff wavelength (μm)
η	efficiency (–)
μ	viscosity (kg m^{-1} s^{-1})
ρ	density (kg m^{-3})
σ_v	volumetric scattering coefficient (m^{-1})
ϕ	porosity (–)
Φ	scattering phase function (–)
Ω	Albedo ($=\sigma_v \beta^{-1}$)

Subscripts

a	aperture
b	back
ex	exit
ext	external
f	fluid
in	inlet
s	solid
v	volumetric
LW	long wave spectral band, $\lambda > \lambda_c$
SW	short wave spectral band, $\lambda < \lambda_c$

Conventional Combined Cycles can reach high conversion efficiency of around 60% from heat to electricity, that would correspond to overall solar to electricity conversion efficiency of 25–30% (Kribus et al., 1998). A Combined Cycle requires heating compressed air to temperatures above 1000 °C. Providing solar heat at these temperatures faces significant challenges, including materials for high temperature, optimization of radiative and convective heat transfer in the receiver, optics for high concentration, and adapting gas turbines for solar or hybrid operation. Some major advances were achieved at the lab level for high-temperature receivers, for example heating air at 20 bar to 1200 °C (Kribus et al., 2001). However, most of the R&D work done in recent years focused on lower-temperature versions of air-heating receivers, intended for indirect steam generation with air at temperature of around 700 °C (Hoffschmidt et al., 2003), or for simple cycle gas turbine plants with solar air preheating to about 800 °C (Schwarzbözl et al., 2006). These lower temperature applications can be viewed as intermediate steps in the long-term vision, not yet reaching the highest possible efficiency, but providing valuable experience with solar air heating technologies.

A key component in the solar thermal conversion process is the radiation absorber located at the focus of the concentrator field. For high temperature, a porous volumetric absorber should provide higher efficiency compared to a tubular receiver, due to the so-called "volumetric effect" (Ávila-Marín, 2011). This effect is defined as the existence of low absorber temperature at the front side of the absorber, such that the loss due to emission of radiation to the environment is reduced. The volumetric effect is expected due to the convective cooling of the front of the absorber by the incoming cold air. However, volumetric absorbers tested to date do not show the expected effect: they produce high temperature at the front face and their efficiency is usually around 70%, even for air temperature well below 1000 °C. For example the TSA receiver used a wire-knit volumetric absorber, with reported air exit temperatures up to 780 °C and efficiency around 70% (Tyner et al., 1996). The ceramic grid HITREC absorber reported efficiency around 76% at 700 °C, and 72% at 800 °C (Hoffschmidt et al., 2003). The low efficiency is related to the high temperatures found at the absorber front surface, similar to or even higher than the air outlet temperature (Hoffschmidt et al., 2003; Fend et al., 2004), indicating that the 'volumetric effect' was not achieved. This has also been confirmed in detailed simulations of volumetric absorbers (Wu et al., 2011), as shown in Fig. 1. Higher efficiency in the range 80–90% at air exit temperature of >1100 °C were reported (Kribus et al., 2001) but this was under very high incident radiation flux of several thousand suns, which is impractical for commercial-scale central receiver plants. High efficiency was also reported with a high-temperature

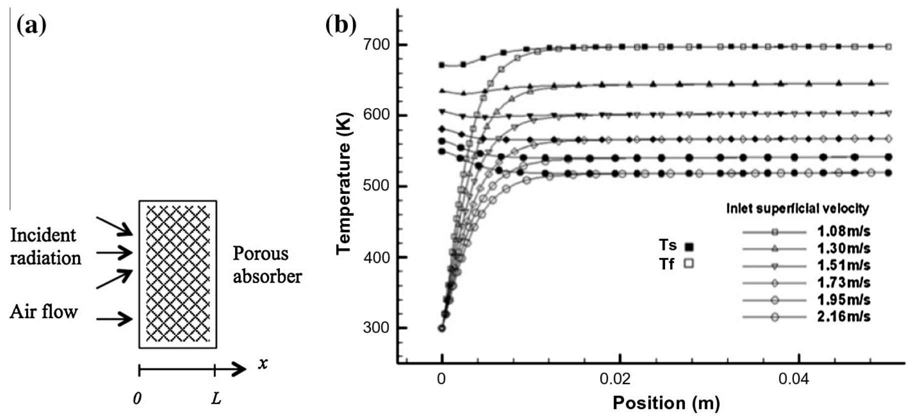

Fig. 1. (a) Layout of the one-dimensional volumetric absorber model, (b) temperature profiles in a volumetric absorber showing high front surface temperature, from a detailed simulation (Wu et al., 2011).

tubular absorber (Hischier, 2012) but again under very high incident radiation flux. Therefore, the promise of volumetric absorbers to provide a high-efficiency path to solar gas turbine and combined cycle plants is yet unfulfilled.

In addition to efficiency, other issues need to be resolved for practical implementation of volumetric receivers. The demanding conditions of high temperature, thermal shock and frequent thermal cycling restrict the range of available materials for the absorber. When the receiver is integrated directly in a gas turbine cycle, the need to maintain air pressure requires the addition of an expensive and fragile high-pressure window (Karni et al., 1998). These additional issues, as well as the cycle configuration and economic aspects, are not addressed in the current work.

This work investigates what is the maximum thermal efficiency that a volumetric absorber may reach under a reasonable incident radiation flux, and whether a significant volumetric effect can in fact be achieved. A significant volumetric effect is defined qualitatively here as a front surface temperature of the absorber that is significantly lower, i.e., by hundreds of degrees, than the fluid exit temperature. A model of the absorber that assumes local thermal equilibrium (LTE, solid and fluid temperatures are equal at each location, equivalent to infinite convective heat transfer coefficient) will produce the volumetric effect by definition. Such a model does predict very high absorber efficiencies, much higher than those achieved in practice (Spirkl et al., 1997). For example, for an absorber operating under incident flux of 800 kW/m^2 with exit air temperature of 1000 °C the predicted efficiency for an ideal LTE absorber is over 95%. Clearly such a simplified model is too optimistic, and a more detailed analysis is needed, taking into account local thermal non-equilibrium (LTNE) as well as more realistic properties of volumetric absorbers.

Various numerical models were developed to solve the coupled flow and heat transfer problem in porous volumetric absorbers, mostly based on volume averaging to represent the porous absorber as a homogeneous medium with effective properties. Cheng et al. (2013) simulated the volume-averaged absorber with a general CFD tool and a Discrete Ordinates (DO) model for radiative transport, but used the LTE assumption that eliminates the distinction between solid and fluid temperatures. Wang et al. (2014) used an averaged medium model with LTNE, and the radiative transport was solved with the Rosseland and P_1 approximations. The Rosseland approximation is appropriate when modeling an optically thick medium; however, the main challenge in the volumetric absorber problem is to describe accurately the front layer where the incident radiation is gradually absorbed and the fluid and solid temperatures change sharply. This layer is not optically thick, even when the absorber as a whole is thick; assuming an optically thick medium eliminates any ability to resolve this layer. The P_1 method is known to give good results for highly scattering media, and at locations away from the boundaries. However, the volumetric absorber problem has the opposite requirements: the crucial process occurs near the front boundary, and the medium is only weakly scattering. Becker et al. (2006) used a finite volume simulation where the radiative transport was simplified into an effective thermal conductivity in the solid phase energy balance. This is similar to the Rosseland approximation mentioned above, setting the absorber to act as an optically thick medium. Wu et al. (2011) solved the LTNE problem but used the modified P_1 approximation for radiative transport (Modest, 2003). In this model, the incident solar radiation is decoupled and solved separately since this part of the radiation breaks the directional distribution assumed in the P_1 model; however, close to the boundary, this method overestimates the radiative losses. Some simpler models were also proposed: for example (Bai, 2010) was able to obtain an analytic one-dimensional solution for the energy balance of the absorber, but at the expense of assuming LTE and representing the radiative transport as a fixed heat source. Therefore, both convective and radiative transport modes were not well represented. Pitz-Paal et al. (1997) solved a coupled set of 1-D problems with LTNE and DO radiative model, which represented different sections of the absorber. However, they simplified the boundary conditions and the material behavior (isotropic scattering), and did not use the model to optimize the absorber structure or study the impact of the different absorber design parameters.

Many of these previous analyses focused on specific samples of porous ceramics and analyzed their performance as volumetric absorbers. Some results were more general, for example the conclusion that higher porosity improves the absorber efficiency. In the LTNE studies it was found that the volumetric effect, as defined above, does not exist: the solid absorber temperature near the aperture is either higher than, or very close to, the temperatures that are found deeper in the absorber (Wu et al., 2011). Variations of absorber porosity, pore size, surface emittance, and flow rate have some effect on the temperature distribution and the absorber efficiency, but are not effective enough to create a significant volumetric effect (Wang et al., 2013). It is interesting to note that some solutions were presented showing a small decrease in absorber temperature at the front (Pitz-Paal et al., 1997), but this was attributed to lateral heat transfer from the center to the rim of the absorber, rather than to the volumetric effect. The only solutions showing low temperature at the front of the absorber are those that use LTE and therefore force artificially the low temperature via the inlet boundary condition, for example (Roldán et al., 2014).

None of the aforementioned models has shown how the desirable volumetric effect might be achieved, or whether it can be achieved at all, in a real absorber. Consequently, the theoretical upper limit on absorber efficiency is not known. The current work aims to address these questions. The model used here is formulated with the intention to be realistic enough in representing the most significant physical processes, yet simple enough to perform many fast computations for an extensive parametric study. We focus specifically on modeling of absorbers made of open-cell ceramic foams (reticulated porous ceramics, RPC), although it should be noted that other geometries and materials have been proposed for volumetric absorbers (Ávila-Marín, 2011). The analysis developed here includes coupled modeling of all three modes of heat transfer, with careful consideration of realistic boundary conditions and of appropriate correlations for effective material properties. A previous version of this analysis (Kribus et al., 2013) has addressed the same question of performance limits, and the current work presents several major advances: a more accurate radiative model (discrete ordinates); a more realistic convective heat transfer correlation; and a more comprehensive presentation of trends in parameter space rather than just a few example cases. The absorber performance is investigated for a range of design parameters that define geometry and material properties, and the combinations of parameters that can lead to volumetric behavior are identified and evaluated.

2. Model

2.1. Momentum and thermal energy transport

A one-dimensional problem is assumed with changes occurring only in the axial direction perpendicular to the absorber surface, where the flow and radiation transport occur both along this direction, as shown in Fig. 1. The volumetric absorber is modeled as a homogeneous effective medium comprising two continuous phases, solid (porous absorber) and fluid (air). This is a common approach that constructs the model by averaging the transport equations on a representative elementary volume, which is larger than the pore scale but smaller than the problem scale (Quintard et al., 2005). This model is reasonable for structures such as ceramic foam but not for ordered structures such as a honeycomb. The solid and fluid phases have separate temperature distributions $T_s(x)$ and $T_f(x)$, consistent with thermal non-equilibrium. The fluid mass flow rate is constant through each cross section due to the 1-D assumption and therefore the mass conservation equation is not needed. The momentum balance is represented with a Darcy–Forchheimer type correlation for the pressure drop in the porous medium along the flow in the axial direction (Wu et al., 2010). The thermal energy balance is applied separately to the solid and fluid phases. For the fluid, the transport by conduction is neglected, assuming that it is significantly smaller than the transport by advection. The effect of thermal dispersion, which is equivalent to an increase in the fluid thermal conductivity, is not explicitly modeled here, but it is indirectly included in the correlation for the volumetric heat transfer coefficient. The fluid is completely transparent to the radiation. For the solid, the area for conduction is determined by the porosity, and the interaction with the radiation (both incident from outside and emitted by the absorber) is included as the divergence of the radiative heat flux q_R.

$$\dot{m}C_p\frac{dT_f}{dx} = h_v\left[T_s(x) - T_f(x)\right] \tag{1}$$

$$0 = (1-\phi)k_s\frac{d^2T_s}{dx^2} - h_v\left[T_s(x) - T_f(x)\right] - \frac{dq_R}{dx} \tag{2}$$

Inter-phase transport, or exchange of heat between the solid absorber and the air, is represented in the heat balance equations by an averaged convection coefficient per unit volume, h_v. The volumetric convection coefficient includes the effect of specific surface area per unit volume, and is derived from a correlation by Fu et al. (1998) as a function of pore size, the thickness of the absorber, and the Reynolds number based on the pore size:

$$Nu_v = \left[0.0426 + 1.236\left(\frac{d_m}{L}\right)\right]Re_{d_m} \tag{3}$$

The correlation provides an average value over the entire absorber thickness, without any information on spatial variations. Therefore, the average air temperatures between inlet and exit was used to evaluate the fluid properties, and the same convection coefficient was used throughout the absorber, except for the front boundary condition as described below. Further discussion of this correlation, and a comparison to other available correlations and experimental results, is found in Appendix A.

The properties of the air can change considerably over the absorber, due to the large variation in temperature. Variable properties (density, viscosity, specific heat, thermal conductivity) were used, based on fit to tabulated data for air taken from: *http://www.engineeringtoolbox.com*.

2.2. Radiative transport

The volumetric absorber is modeled as a plane slab of a homogeneous participating medium for the radiative transport balance. Three approximate methods to solve the radiative transfer equation were investigated, and compared against a detailed 3-D Monte-Carlo simulation.

2.2.1. Radiative properties

The volumetric extinction, absorption and scattering coefficients for the effective participating medium are defined as:

$$\beta = \frac{C(1-\phi)}{d} \qquad \alpha_v = \alpha \cdot \beta, \quad \sigma_v = (1-\alpha)\beta \tag{4}$$

The value of C was set to 4.8 based on a fit to experimental measurements (Hendricks and Howell, 1996). Further discussion of the correlation and the selected value of C appears in Appendix A. For the Monte Carlo and Discrete Ordinates methods, the scattering phase function was set to an analytic expression (Hendricks and Howell, 1996) that shows a good fit to detailed analysis of a real ceramic foam (Petrasch et al., 2007):

$$\Phi = \frac{8}{3\pi}(\sin\theta - \theta \cdot \cos\theta) \tag{5}$$

θ is the scattering angle defined as the angle between the incident and scattered directions.

2.2.2. Two-flux approximation

The two-flux approximation is based on a minimal discretization of the directional space into two hemispheres, forward and backward, in order to keep the computational effort to a minimum. Assuming isotropic scattering, the approximation leads to two equations for the forward and backward intensities, i^+ and i^- (Modest, 2003):

$$\frac{di^+}{dx} = -2\beta i^+(x) + \beta\Omega(i^+(x) + i^-(x)) + 2\beta(1-\Omega)\phi e_b(x)/\pi \tag{6}$$

$$-\frac{di^-}{dx} = -2\beta i^-(x) + \beta\Omega(i^+(x) + i^-(x)) + 2\beta(1-\Omega)\phi e_b(x)/\pi \tag{7}$$

$e_b(x)$ is the blackbody emission flux at the local temperature $T_s(x)$, and Ω is the scattering albedo. Note that the emission term is multiplied by ϕ, to represent the fact that the internal emission in the bulk absorber propagates in the void space only, not in the entire volume (Taine et al., 2010).

When collimated sunlight is incident at $x = 0$, it is possible to split the intensity into diffuse and collimated components. The intensities i^+ and i^- in Eqs. (6) and (7) then describe only the diffuse component. The collimated flux decays exponentially: $q_c(x) = q_c(0)e^{-\beta x}$. The contribution of the collimated component to the in-scattering source of the diffuse component is: $\beta\Omega q_c(0)e^{-\beta x}/2\pi$, and it should be added in Eqs. (6) and (7).

Given the intensities, the net radiative flux is $q_R = \pi(i^+ - i^-)$, and the net radiative heat source per unit volume absorbed by the medium that is needed for Eq. (2) is given by the divergence of the radiative flux. Combining (6) and (7), the scattering terms are canceled out and we obtain:

$$-\frac{dq_R}{dx} = 2\pi\alpha_v(i^+(x) + i^-(x)) - 4\alpha_v\phi e_b(x) \tag{8}$$

When a collimated flux component is present, it should also be added in the Eq. (8).

Eqs. (6)–(9) refer to a gray medium. The treatment is extended to spectrally selective absorbers by defining two intensities in each direction $i^\pm_{SW}$, $i^\pm_{LW}$ to represent two spectral bands, separated by wavelength λ_c. The set of radiative transport Eqs. (6) and (7) is applied separately in each of the two spectral bands, where the optical properties are different in each band. The emission term is divided to the spectral bands using the fractional emissive power function $F(\lambda T)$ and it complement $\overline{F}(\lambda T) = 1 - F(\lambda T)$. The net radiative heat source then includes the contributions of the two spectral bands:

$$-\frac{dq_R}{dx} = 2\pi\alpha_{v,SW}\left(i^+_{SW}(x) + i^-_{SW}(x)\right) + 2\pi\alpha_{v,LW}\left(i^+_{LW}(x) + i^-_{LW}(x)\right) - 4\phi e_b(x)\left(\alpha_{v,SW}F(\lambda_c T_s(x)) + \alpha_{v,LW}\overline{F}(\lambda_c T_s(x))\right) \tag{9}$$

2.2.3. S_4 approximation

The discrete ordinates (DO) method was implemented using a low-order S_4 formulation with 24 ordinates (Modest, 2003). Due to axial symmetry only four ordinates need to be solved, two directions in the forward hemisphere i^+_j and two backward directions i^-_j, where $j = 1, 2$. The DO method should allow a better representation of the directional distribution of incident radiation and radiation within the participating media, compared to the two-flux method. The radiative transfer equations for the forward and backward directions are:

$$\frac{\mu_j}{\beta} \cdot \frac{di^+_j}{dx} = (1-\Omega)\frac{\phi e_b(x)}{\pi} - i^+_j(x) + \frac{\Omega}{4\pi}\sum_{k=1}^{24} w_k\left[i^+_k(x) + i^-_k(x)\right]\Phi_{jk} \tag{10}$$

$$-\frac{\mu_j}{\beta} \cdot \frac{di^-_j}{dx} = (1-\Omega)\frac{\phi e_b(x)}{\pi} - i^-_j(x) + \frac{\Omega}{4\pi}\sum_{k=1}^{24} w_k\left[i^+_k(x) + i^-_k(x)\right]\Phi_{jk} \tag{11}$$

μ_j is the polar angle cosine for ordinate j ($j = 1,2$), and w_k is the quadrature weight of ordinate k ($k = 1,\ldots,24$) (Modest, 2003). Φ_{jk} is the scattering phase function evaluated from Eq. (5) between incident ordinate j and scattered ordinate k. The radiative heat source is:

$$-\frac{dq_R}{dx} = \sum_{j=1}^{2} w'_j \mu_j \left(\frac{di_j^+}{dx} - \frac{di_j^-}{dx} \right) \tag{12}$$

When solving for a non-gray medium in the two spectral bands, each ordinate direction corresponds to two intensities $i^{\pm}_{j,SW}$, $i^{\pm}_{j,LW}$. Eqs. (10) and (11) are duplicated for the two spectral bands, and the radiative heat source is then:

$$-\frac{dq_R}{dx} = \sum_{j=1}^{2} w'_j \mu_j \left[\frac{di^+_{j,SW}}{dx} + \frac{di^+_{j,LW}}{dx} - \frac{di^-_{j,SW}}{dx} - \frac{di^-_{j,LW}}{dx} \right] \tag{13}$$

2.2.4. P_1 model

This radiative model comes from the method of spherical harmonics, P_N, using the first two terms of the spherical harmonics series of the radiation intensity, $i^{(0)} = G$ (total incident radiation) and $i^{(1)} = q_R$ (radiative flux) (Modest, 2003). Although this method can be used with different scattering phase functions, we have used it here only with isotropic scattering. The closure condition for the P_1 model in this case leads to two differential equations:

$$\begin{cases} \frac{dq_R}{dx} = \alpha_v (4e_b(x) - G) \\ \frac{1}{3\beta}\frac{dG}{dx} = -q_R \end{cases} \tag{14}$$

This can be formulated as a single diffusion equation for the total incident radiation (direction-integrated intensity) G:

$$\begin{aligned} -\frac{d}{dx}\left(\frac{1}{3\beta}\frac{dG}{dx}\right) &= \alpha_v (4e_b(x) - G) \\ &= \alpha_v \left(4\sigma_{sb} T_s^4(x) - G\right) \end{aligned} \tag{15}$$

After solving for the total incident radiation, the gradient of the radiative flux is computed using Eq. (14), leading to the radiative source term needed for the solid phase energy equation, Eq. (2).

As presented in section 2.2.2, if incident collimated radiation is present, it can be treated separately. The total irradiance is the sum of diffuse (scattered and emitted) and collimated contributions $G = G_d + G_c$, where the collimated irradiance is $G_c(x) = q_c(0)e^{-\beta x}$, and $q_c(0)$ is the collimated flux entering the medium. Eq. (15) then solves for the diffuse irradiance G_d.

2.2.5. Monte Carlo method

A Monte Carlo (MC) method (Delatorre et al., 2014) was implemented for reference computations. The MC algorithm solves the integral expressions of the radiative source term and the radiative losses of the porous medium. The MC algorithm includes the intensity splitting technique and the importance sampling technique to sample the incident and scattering directions. The optical path is randomly generated with a probability density function depending on the scattering coefficients. The absorption is accounted along this random path. Two MC algorithms were used to separate the contribution of the external conical solar source and the contribution of thermal emission inside the medium.

2.3. Boundary conditions

2.3.1. Front boundary $x = 0$

The front aperture of the absorber is modeled as a pseudo-surface that interacts with the radiation and with the bulk absorber behind it. The surface has a porosity ϕ and surface absorptance α, which are assumed to be the same as the properties inside the absorber. The solid fraction of the surface, $1 - \phi$, absorbs the radiation incident on it, leading to local heating at the surface. The aperture surface boundary condition is then a balance of radiation, conduction to the solid bulk absorber, and convection to the entering fluid:

$$\alpha q_{in} = \alpha\sigma T_s^4(0) - k_s \left.\frac{dT_s}{dx}\right|_0 + h_a \left(T_s(0) - \frac{T_f(0) + T_{in}}{2} \right) \tag{16}$$

The temperature of the fluid is taken as the average of the inlet state upstream of the absorber, and the point of entry into the absorber after exchanging heat with the front surface. The surface convection coefficient at the aperture h_a is based on the analysis in Wu et al. (2011) that provides the spatial variation of convection coefficient in the entrance region of a reticulated structure. The surface convection coefficient in the developed region is calculated from the volumetric coefficient, Eq. (3), and the specific surface area; the surface convection coefficient at the aperture is then set to 1.7 times the developed value, based on the same ratio found in Wu et al. (2011). The corresponding boundary condition for the fluid phase, representing the convection of heat from the front pseudo-surface before the fluid enters the bulk absorber, is:

$$\dot{m}C_p\left(T_f(0) - T_{in}\right) = (1-\phi)h_a \left(T_s(0) - \frac{T_f(0) + T_{in}}{2} \right) \tag{17}$$

The simultaneous solution of (16) and (17) defines $T_f(0)$ and $T_s(0)$, where T_{in} is given as input. Due to the dependence of h_a on the temperature, this is solved iteratively. The radiation flux entering the bulk of the absorber, after a partial interception by the frontal area, is ϕq_{in}. The boundary condition for the two-flux approximation is then:

$$i^+(0) = \phi q_{in}/\pi \tag{18}$$

For the P_1 model, the inlet boundary condition taking into account the solar energy entering the medium after partial interception by the frontal area is:

$$-\frac{1}{3\beta}\left.\frac{dG}{dx}\right|_0 = \phi q_{in} - \frac{G(0)}{2} \tag{19}$$

$G(0)/2$ is the expression of the radiative losses towards the ambient, obtained when the Marshak boundary condition is used (Modest, 2003). For the S_4 approximation we apply at the aperture two radiative boundary conditions to define the two incident ordinates. The first is the incident heat flux:

$$\sum_{j=1}^{2} \mu_j w'_j i_j^+(0) = \phi q_{in} \tag{20}$$

w'_j is the summed quadrature weight of ordinate j (Modest, 2003), taking into account the sum of all identical ordinates due to axial symmetry. The second boundary condition is based on the second moment of the incident radiation intensity:

$$\int_{\varphi=0,\theta=0}^{\varphi=2\pi,\theta=\frac{\pi}{2}}\int i_{in}(\theta)\cos^2\theta \sin\theta d\theta d\varphi = \sum_{j=1}^{2} \mu_j^2 w'_j i_j^+(0) \tag{21}$$

i_{in} is the intensity directional distribution created by the solar concentrator. In this work we assume that the incident radiation intensity is uniform within a cone of opening angle θ_0, and zero outside this cone. The contribution of ambient diffuse radiation is ignored since in most cases it is negligible compared to the concentrated direct radiation incident flux.

$$i_{in}(\theta) = \begin{cases} \frac{q_{in}}{\pi sin^2\theta_0}; & 0 \le \theta \le \theta_0 \\ 0; & \theta > \theta_0 \end{cases} \tag{22}$$

Lambertian incident radiation is obtained when $\theta_0 = \pi/2$, and collimated incident radiation corresponds to the limit $\theta_0 \to 0$.

2.3.2. Back boundary $x = L$

The back surface of the absorber at $x = L$ is also defined as a pseudo-surface that is partly solid, with a surface porosity that is assumed equal to the volumetric porosity ϕ. An empty space is provided behind the absorber to allow exit and collection of the hot fluid, enclosed with a back wall, as shown in Fig. 2. A radiative enclosure is defined between two parallel planes, where surfaces 1 and 2 are the solid and void parts of the back surface of the absorber, and 3 represents the back wall. In case of a more complex geometry of the fluid exit duct, the back wall is a virtual surface that represents the duct's effective radiative properties. Given the properties of the back wall, the radiative exchange in the enclosure can be solved, and the resulting net flux absorbed at the back wall is the energy loss from the back of the absorber to the environment. For example, if the back wall is perfectly reflective, then the loss at the back of the absorber is zero.

Consistent with the one-dimensional nature of the model, the sidewalls of the cavity are assumed to have negligible influence, and surfaces 1, 2 and 3 exchange radiation essentially as infinite parallel planes. The enclosure radiative problem follows the standard formulation (Modest, 2003) for the incident and outgoing radiative fluxes (irradiance and radiosity) $q_{in,i}$, $q_{out,i}$ for each surface i $(i = 1, 2, 3)$:

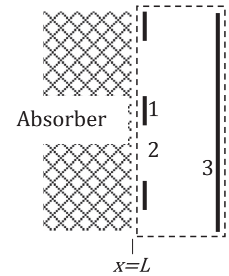

Fig. 2. The radiative problem at the end of the absorber: dashed line is the radiative enclosure; 1 solid absorber edge; 2 void; 3 back wall.

$$\begin{aligned} q_{out,i} &= \alpha_i \sigma T_i^4 + (1-\alpha_i) q_{in,i} + q_{ext,i} \\ q_{in,i} &= \sum_{j=1}^{3} F_{i-j} q_{out,j} \end{aligned} \tag{23}$$

α_i is the surface absorptance/emittance of surface i. Surface 1 has emittance equal to the absorber material's surface absorptance, and its temperature is $T_s(L)$. Surface 2 has emittance 1 (incident radiation is not reflected), and its temperature is set to zero, since there is no surface emission, only the specified external flux $q_{ext,2}$ that enters the enclosure from the absorber. The emittance and temperature of the back wall depend on the receiver design.

$q_{ext,i}$ is the external radiation flux that enters the enclosure through surface i. For the opaque surfaces 1 and 3 it is zero, and for the virtual surface 2 it is radiation flux entering from the bulk of the absorber. For the 2F method it is: $\pi i_1^+(L)/\phi$. For the discrete ordinate solution, the external flux is obtained from the intensities of the S_4 solution in the bulk absorber:

$$q_{ext,2} = \frac{\left[\mu_1 w'_1 i_1^+(L) + \mu_2 w'_2 i_2^+(L)\right]}{\phi} \tag{24}$$

The division by the porosity represents the difference in areas between the computational homogeneous medium domain inside of the absorber, and the real area where radiation can exit the back surface of the absorber. The view factors between the 3 surfaces are: $F_{1-3} = F_{2-3} = 1$; $F_{3-1} = 1 - \phi$; $F_{3-2} = \phi$; all other view factors are zero. Given a full set of boundary conditions, the enclosure problem can be solved numerically or analytically to find the incident and outgoing fluxes on all surfaces.

The net flux that is lost via the back wall is $-q_{net,3} = q_{in,3} - q_{out,3}$. The boundary condition for the solid phase of the edge of the absorber, considering conduction through the solid absorber, convection at the edge surface, and radiative exchange with the back enclosure, is then:

$$-k_s \frac{dT_s}{dx}\bigg]_L = q_{net,1} + h_b\left(T_s(L) - \frac{T_f(L) + T_{ex}}{2}\right) = 0 \tag{25}$$

The convection coefficient at the back surface h_b is taken from (Fu et al., 1998). The fluid exit temperature T_{ex} is calculated from an energy balance on the exiting fluid:

$$\dot{m}C_p\left(T_{ex} - T_f(L)\right) = (1-\phi)h_b\left(T_s(L) - \frac{T_f(L) + T_{ex}}{2}\right) \tag{26}$$

The boundary condition for the backwards intensity (2F model) or the two negative intensities (S_4 model) in the bulk absorber is derived from the net flux of surface 2, under the assumption that radiation on all surfaces in the enclosure is diffuse:

$$i_1^-(L) = i_2^-(L) = \phi\frac{q_{in,2}}{\pi} \tag{27}$$

2.4. Absorber performance

The set of differential equations and boundary conditions was solved using MATLAB. After the solution is obtained, the absorber efficiency can be calculated as the thermal power per unit area imparted to the fluid, divided by the incident flux:

$$\eta = \dot{m}\int_{T_{in}}^{T_{ex}} C_p(T)dT/q_{in} \tag{28}$$

The different losses can be derived separately from the solution. When the back surface of the absorber is assumed to be a perfectly reflector, there are no losses at the back surface. The losses related to the front pseudo-surface include reflection and thermal emission:

$$(1-\phi)\left[(1-\alpha_a)q_{in} + \alpha_a\sigma T_s^4(0)\right] \tag{29}$$

The loss by back scattering and emission from the bulk of the absorber is $\pi i^-(0)$ for the two-flux solution, and the corresponding expression for the S_4 solution is:

$$\sum_{j=1}^{2}\mu_j w_j' i_j^-(0) \tag{30}$$

It is possible to separate the two contributions by repeating the simulation with a cold medium ($T_s(x) \approx 0$ K), a case that produces only back scattering loss. This can be subtracted from the full result of Eq. (30) to obtain the bulk emission loss.

2.5. Validation

A major uncertainty in the model is the use of the very simple approximations to represent the radiative transport in the absorber. The validity of these approximations was tested by comparison to the MC simulation. Two numerical validation cases (Table 1) were chosen to compare the accuracy of the radiative transfer methods: the first one is a cold medium (only radiative transport is solved, and the problem is decoupled from the convective energy balance) with a single scattering albedo of 0.9, and isotropic scattering. The second case is a hot medium with realistic radiative properties of ceramic foam volumetric absorbers, where fully coupled simulations are run with the MC and DO radiative solution. The directional extent of the incident radiation for both MC and DO simulations was defined a cone with opening half-angle of 45°. The two-flux approximation was computed using the two extreme cases of collimated and diffused incident radiation, and the P_1 model used only collimated incident light.

Fig. 3 shows the total irradiance inside the 1-D absorber for the cold medium case, using the different radiative transfer methods and corresponding incidence. The MC simulations needed 10^6 bundles of rays to compute the incident radiation and the boundary fluxes due to the solar source, whereas 5×10^4 bundles were needed to compute the radiation and the boundary fluxes due to the medium thermal emission. These numbers of bundles were used to obtain a standard deviation associated to the MC estimate lower than 0.2%. The maximum relative error for the DO (S_4) is 5.1% compared to the MC results. The small discrepancies in the irradiance profile should be due to the limited number of directions used with the S_4. The 2F approximation with a diffuse incident flux overestimates the irradiance at the entrance, whereas both the 2F and P_1 models underestimate the irradiance when a collimated source is considered. Indeed, the P_1 and 2F methods are not able

Table 1
Absorber properties for cold and hot validation cases.

	Cold simulation	Hot simulation
Absorptance	0.097	0.90
Absorption coefficient (m^{-1})	15	108 (Eq. (4))
Scattering coefficient (m^{-1})	140	12 (Eq. (4))
Scattering phase function	Isotropic	Eq. (5)
Thermal conductivity ($W\,m^{-1}\,K^{-1}$)	–	1.0
Mass flow per unit area ($kg\,s^{-1}\,m^{-2}$)	–	0.65
Porosity	0.90	0.90
Mean cell size (mm)	4	4
Incoming solar power ($kW\,m^{-2}$)	800	800

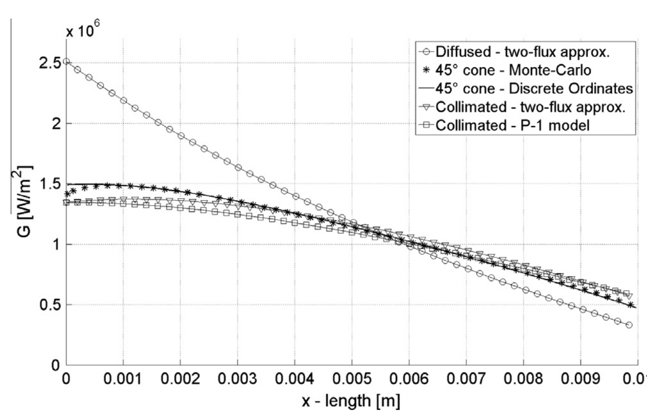

Fig. 3. Cold medium case- comparison of Monte-Carlo, Discrete Ordinates S_4, Two-flux approximation and P_1 model total irradiance for a highly scattering cold medium.

to handle the conical incidence, and forcing a diffuse or collimated incidence leads to significant differences. Based on this first numerical test, we choose the S_4 as the best method to model the radiative transfer inside volumetric absorbers, and the second numerical comparison will focus on S_4 and MC.

Fig. 4 presents the radiative source terms and the temperature fields inside the absorber obtained for the "hot" validation case after convergence of the coupled absorber model using either S_4 or MC. Small discrepancies on the radiative source term are observed with a relative global error of 0.82%. Those differences are due to the inlet boundary condition. In the case of the Monte-Carlo, all the flux is in a 45° cone leading to a deeper penetration inside the absorber (higher radiative source term inside the absorber). Concerning the Discrete Ordinates, the flux is spread on the two positive intensities, meaning that there is flux inside the whole positive hemisphere (not only a cone). Thus, the radiative source term is slightly lower.

The solid and fluid temperature fields, Fig. 4(b), present a maximum temperature difference of 13 K for the solid. Similarly, a negligible difference of 0.2% is found for the solar-to-thermal efficiency, due to the difference in fluid temperature at the outlet (around 8 K).

A comparison between the computing time of the S_4 and MC was done. The convergence of the fully coupled absorber simulation needs almost 6 h with a parallelized MC (8 processors), equivalent to about 170,120 s for one processor. For the same simulation but solving the radiative heat transfer with the S_4 method, the full convergence is obtained in less than 12 s with 1 processor. The use of S_4 leads then to a time reduction factor of more than fourteen thousands (about 14,200). Consequently, we conclude that the S_4 approximation is very efficient in computation time, and accurate enough to describe the behavior of radiative transfers inside a volumetric absorber. Therefore, its use is confirmed for the remainder of the study.

3. Results

3.1. Parametric study setup

A set of absorber data was selected as the baseline for the investigation of volumetric absorber performance, as shown in Table 2. This corresponds to an open receiver made of SiC foam, where air enters at atmospheric pressure and ambient temperature, and the incident radiation is produced with a high quality heliostat field with little or no secondary concentration. The mass flow rate was selected based on the estimate that if the absorber efficiency would be around 90% under the given incident flux, then the air exit temperature would be about 1000 °C.

All of the absorber geometric, optical and thermal properties are set to a uniform value across the absorber. The possibility of graded properties (changing over the depth of the absorber) was not considered in the scope of the current work, but it should be investigated as another possible means to influence the absorber performance. A preliminary study was done for the thermal conductivity using temperature-dependent conductivity of a SiC absorber vs. a constant average value, showing that the average value is a good approximation. The surface emittance of the front surface is the same as the bulk material. The reflectance of the back wall of the receiver is 1, indicating that there is no radiative energy loss at the back of the absorber.

Table 2
Volumetric absorber data common to all cases.

Incident flux	800 kW m^{-2}
Air inlet temperature	300 K
Air inlet pressure	10^5 Pa
Mass flow rate per unit area	0.65 kg s^{-1} m^{-2}
Absorber thickness	0.020 m
Absorber thermal conductivity	40 W K^{-1} m^{-1}
Absorber surface emittance (gray)	0.9

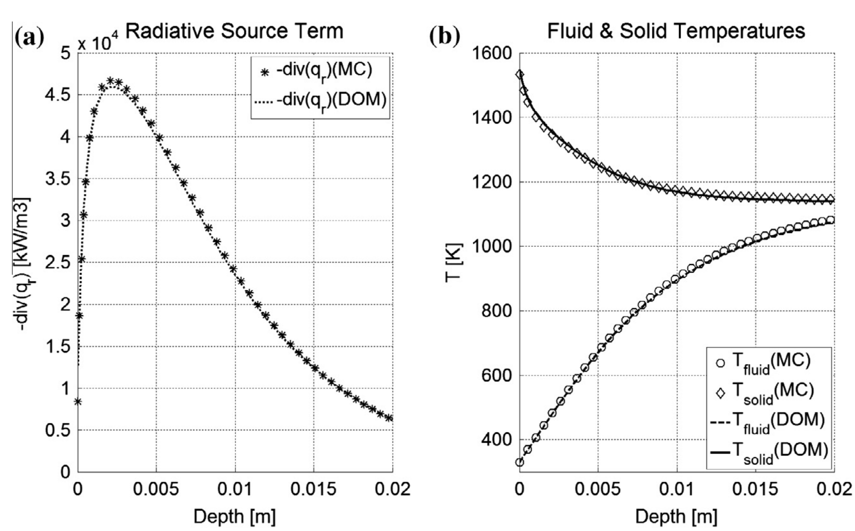

Fig. 4. Comparison of the "hot" case with the Discrete Ordinates (S_4 approximation) and the Monte-Carlo simulations, (a) Radiative source term, and (b) temperatures profiles.

3.2. Effect of geometric parameters

The impact of porosity and pore size on the absorber performance have been studied previously. For example, higher porosity and smaller pore size seem to improve performance (Wu et al., 2011); however, that analysis lacked a realistic front boundary condition, and relied on a less accurate radiative model, as discussed above. Applying the present model over a range of porosity and pore size values leads to the efficiency results shown in Fig. 5(a). The highest efficiency is found with higher porosity and higher, not lower, pore size. The efficiency is quite low, 0.76 or less over the entire range shown. The temperature distribution over the absorber for the best case ($d = 4$ mm, $\phi = 0.9$), Fig. 5(b), shows that the absorber temperature at the front surface is higher than the air exit temperature, and much higher than the local air temperature near the inlet, hence no volumetric effect is present. The losses are dominated by the high absorber temperature, and include thermal emission from the interior of the absorber (17.2%), and from the solid part of front surface (2.1%).

The poor performance of the absorber can be explained by the fact that the front layers of the absorber have the highest volumetric heat source due to absorption of the incident radiation, both inside the bulk of the absorber, and the localized absorption at the front surface. The mechanism for heat removal- convection to the air, and conduction deeper into the absorber, are insufficient to maintain a lower temperature. The thermal balance is achieved when the absorber temperature is high enough to remove all absorbed radiation by emission and convection. It is clear then that in order to achieve a volumetric behavior and higher efficiency, it is necessary to increase the alternative heat removal mechanisms. The thermal conductivity is already very high for a ceramic material (SiC is one of the best materials for heat conduction). Therefore, it is necessary to significantly enhance convective heat transfer within the absorber.

3.3. Effect of local convection

The means to enhance the convective heat transfer may include for example geometric changes of the pore structure, and modifications of the surface roughness within the pores. It is likely that such changes are difficult to achieve with existing methods of producing ceramic foams, and may require the development of new fabrication technologies. The current study does not address the fabrication methods, but rather investigates what improvement of volumetric absorber performance may be expected, if the volumetric convection coefficient could be increased by a given factor.

Fig. 6(a) shows the improvement in absorber efficiency when the volumetric convection coefficient calculated from Eq. (3) is multiplied by a factor between 2× and 5×. Doubling the convection leads to a significant increase in efficiency from 0.76 to 0.83, while enhancement by a factor of 5 reaches efficiency of 0.86. Also shown are a significant decrease in the front surface absorber temperature $T_s(0)$, and an increase in the air exit temperature $T_f(L)$, due to the enhancement of convective transport. Fig. 6(b) shows the solid absorber temperature profiles $T_s(x)$ for the different values of the convection enhancement factor. For the higher values, a significant volumetric effect can be observed, corresponding to the significantly higher efficiency. The fluid temperature profiles (not shown) do not exhibit a qualitative change, and the fluid temperature increases monotonically in all cases. However, the rate of fluid temperature increase is higher, and the fluid exit temperature is higher, when convection is increased. This corresponds to the higher absorber efficiency as shown in Fig. 6(a).

It is interesting also to note that in the cases with a volumetric effect, the absorber temperature does not increase monotonically, and a small region near $x = 0$ shows higher temperature than further inside. This is a result of the localized absorption at the front surface, as expressed in the boundary condition of Eq. (16). This local effect is relatively small in Fig. 6(b) due to the high porosity (low solid

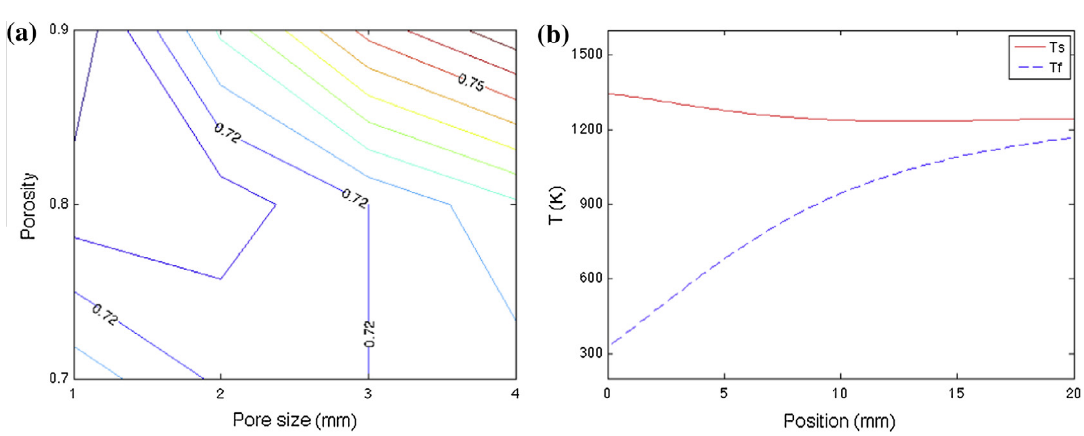

Fig. 5. (a) Absorber efficiency vs. porosity and pore size; (b) temperature distribution for the case $d = 4$ mm, $\phi = 0.9$. Other parameters values in Table 2.

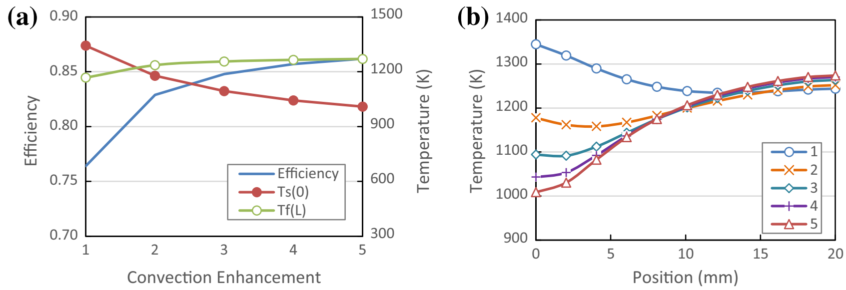

Fig. 6. (a) Effect of convection enhancement on the absorber efficiency and temperatures, for the case $d = 4$ mm, $\phi = 0.9$, and other parameter values from Table 2; (b) absorber temperature distributions $T_s(x)$ for different levels of convection enhancement.

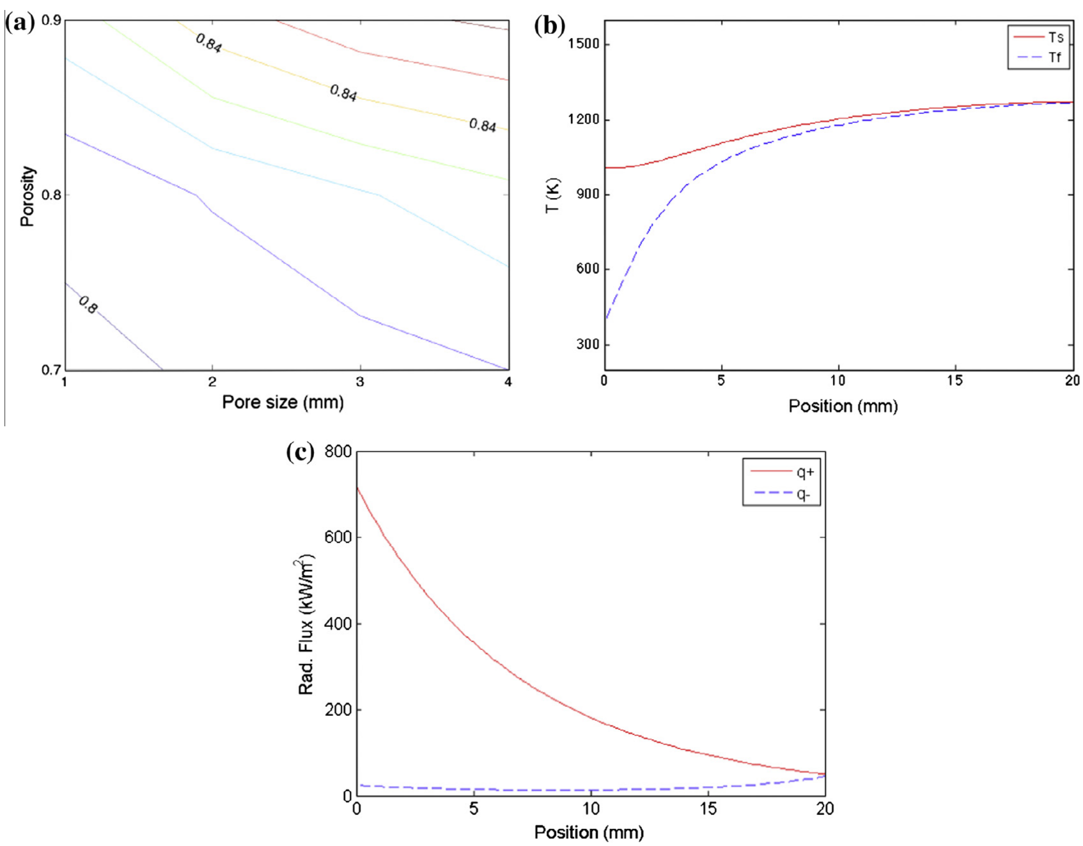

Fig. 7. (a) Absorber efficiency vs. porosity and pore size with 5× enhancement of the convection coefficient, other parameter values are from Table 2; (b) temperature distribution for $d = 4$ mm, $\phi = 0.9$; (c) forward and backward solar radiative flux.

fraction) in this case. For cases with lower porosity, this local 'hot spot' at the front is more significant.

Fig. 7(a) shows the dependence of absorber efficiency on porosity and pore size in the case where the convective heat transfer is enhanced by 5×. The highest efficiency of 0.86 is achieved with higher porosity and larger pore size. Fig. 7(b) shows the temperature distribution across the absorber for the best case with $d = 4$ mm, $\phi = 0.9$, with a distinct volumetric effect ($T_s(0)$ lower than $T_f(L)$ by 261 K. The distribution of the forward and backward radiative flux, as seen in Fig. 7(c), indicates that most of the incident solar radiation is absorbed in the bulk of the absorber and only a small fraction reaches the back end of the absorber. Therefore the thickness of the absorber seems adequate. The back scattering loss that includes both the reflection from the front surface and radiation that is back reflected from the depth of the absorber is only 3.1% of the incident flux. The dominant loss by emission from the bulk of the absorber is reduced to 9.1%, compared to 17.2% in the case of unenhanced convection.

3.4. Effect of thermal conductivity

When a volumetric temperature profile is established, high thermal conductivity enables the undesirable transport of heat by conduction from the back of the absorber towards the front. This diminishes the temperature gradient and can be detrimental to the absorber performance.

A hypothetical case is considered here where the thermal conductivity of the absorber material is set to $k_s = 1$ W m^{-1} K^{-1}, typical of metal oxide materials, but assuming that the high optical absorptance remains unchanged, which is not typical for common oxides. Fig. 8(a) shows the dependence of absorber efficiency on porosity and pore size for the case showed in Fig. 7(a) but with the low absorber thermal conductivity. The highest efficiency rises to 0.88 due to this change, since the low absorber thermal conductivity permits a steeper temperature gradient along the depth of the absorber, increasing the volumetric effect: $T_s(0)$ is lower than $T_f(L)$ by 360 K. Fig. 8(b) shows this stronger volumetric effect, in comparison to Fig. 7(b), achieved due to the lower thermal conductivity of the solid absorber. The gain in efficiency is about 2%, which is not dramatic but still a significant improvement compared to the higher thermal conductivity case.

The local heating effect of the front surface is more pronounced in the case of low thermal conductivity, since heat removal from the front surface by conduction is also reduced, and the front surface temperature is increased until convection and radiation can remove the excess heat absorbed at the front surface. In fact, if the thermal conductivity is reduced even further, this local high temperature can negate the improved volumetric effect and increase the emission loss. This is shown in Fig. 8(c), where conductivity below about 3 W/m K causes an increase the front temperature, and below about 1 W/m K the efficiency shows a slight decrease. This indicates an optimum in the thermal conductivity property, but around this optimum, for 3 W/m K and lower, the variation in efficiency is small.

It is important to note that the effect of low thermal conductivity is helpful in gaining higher efficiency by increasing the volumetric effect, but only when a volumetric effect exists to begin with. Fig. 8(c) shows the impact of low thermal conductivity when no enhancement of convection is present, and therefore no volumetric effect exists. The efficiencies decrease significantly in comparison to Fig. 5, since the low convection along with low solid thermal conductivity result in higher, rather than lower, absorber temperatures at the aperture leading to higher emission loss.

3.5. The effect of spectral selectivity

Spectral selectivity is usually not considered at elevated temperatures, due to the increasing overlap between the solar spectrum and the emission spectrum. Nevertheless, we investigate here this possibility and evaluate its possible impact on the volumetric absorber performance. The simulation of the best case ($d = 4$ mm, $\phi = 0.9$, $k_s = 1$ W m^{-1} K^{-1}, other parameters from Table 2) was repeated for several IR emittance values with a constant absorptance of 0.9. The case an ideal selective absorber with surface emittance of 1 for $\lambda < \lambda_c$ and 0 for $\lambda > \lambda_c$ was also

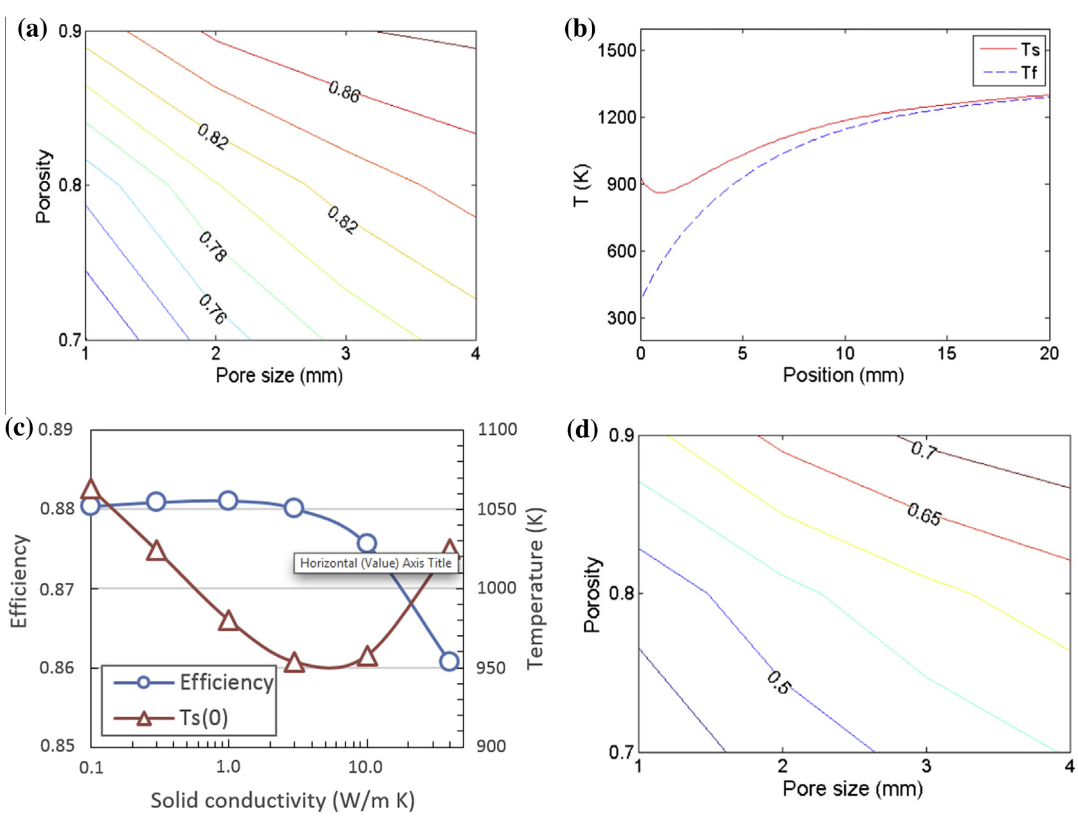

Fig. 8. (a) Absorber efficiency vs. porosity and pore size with absorber thermal conductivity of 1 W m^{-1} K^{-1} and 5× enhancement of the convection coefficient; (b) temperature distribution for $d = 4$ mm, $\phi = 0.9$; (c) efficiency and front solid temperature $T_s(0)$ vs. solid thermal conductivity k_s, for $d = 4$ mm, $\phi = 0.9$ and 5× convection enhancement; (d) absorber efficiency vs. porosity and pore size for $k_s = 1$ W m^{-1} K^{-1} with no enhancement of convection.

included as a theoretical upper limit. The cutoff wavelength λ_c was determined for each case by optimization to yield maximal efficiency.

Fig. 9(a) shows the dependence of the efficiency on IR emittance. The case of an ideal selective absorber together with 5× convection enhancement yields a very high efficiency of 0.97 under the current conditions (given incident radiation flux and air flow rate). For non-ideal spectral selectivity, the improvement from gray (emittance 0.9) to strongly selective (emittance 0.1) increases the absorber efficiency by about 4% to a value of 91%. These results indicate that there is no strict theoretical upper limit on volumetric absorber efficiency; if the materials and means can be found to achieve spectral selectivity and enhanced convection, then the absorber efficiency should reach close to 1.

The temperature distribution for this ideal selective case, Fig. 9(b), shows a good volumetric effect similar to the non-selective case shown in Fig. 8(b). The addition of ideal selectivity then does not change much the nature of the temperature profile, but it does reduce the amount of radiation emitted and lost from the absorber, and the resulting efficiency is then higher.

The case with low thermal conductivity and no convection enhancement was also simulated for several IR emittance values, and the results are also shown in Fig. 9(a). The solid absorber temperatures are much higher, and absorber efficiency is lower, close to the range measured in actual experiments. In spite of the higher temperatures, it is found that spectral selectivity is quite effective. Changing from gray (emittance 0.9) to strongly selective (emittance 0.1) increases the absorber efficiency by 4.3% to a value of 78%, and ideal selectivity in this case yields efficiency of 83.8%. The cutoff wavelength λ_c that is needed in this case shifts to lower values, due to the shift of the emission spectrum to shorter wavelengths at higher temperatures. For example, in the ideal selectivity case, a value of 2.01 μm is needed when no convection enhancement is present, while 2.65 μm is the best value for the same case but with increased convection and lower absorber temperatures.

Finally, it is important to note that real materials with good spectral selectivity at high temperature are not easy to find. Materials are available with high absorptance in the solar spectrum, but their emittance at longer wavelengths will usually be similar, or only somewhat lower. Therefore achieving the full benefit of spectral selectivity will be difficult in practice.

3.6. The effect of front surface absorption

The absorption of radiation at the aperture surface has been largely ignored in past research. In one case it has been discussed but then neglected (Pitz-Paal et al., 1997) with the reasoning that for high porosity ceramic foams the effect of front surface absorption is small and unimportant. In this study we apply a full energy balance at the front surface as described in Eq. (16), and compare the results to the case where this local heat source is neglected as done in the past. Fig. 10 presents this comparison for two cases with low and high convection. In both cases, a local rise in temperature can be observed near the aperture, caused by the absorption of radiation at the aperture. The local temperature rise is about 110 °C in (a) and 100 °C in (b). This is a substantial difference in behavior that should not be neglected. The impact on absorber efficiency is small but not negligible: for the case shown in Fig. 10(a), the efficiency declines by 2.2% when adding the front surface absorption. For the case shown in Fig. 10(b), the decline is by 0.8%, since the temperatures and the related emission losses are smaller. It should be noted that the model used here is not highly accurate but an approximation, compressing the energy transport effects of the front layer (with depth of possibly one or two pore diameters) into a virtual boundary surface of zero depth. Nevertheless, the results shown here indicate that these boundary effects are important and should be represented in modeling and analysis of the absorber.

In addition to the significance of absorber efficiency, one of the few performance characteristics of the absorber that can be acquired during operation is emitted IR radiation exiting from the front surface, which can be measured remotely with an IR camera. When comparing this measurement to simulation, a correct representation of the front boundary condition is essential. The full solution of the front side boundary condition does not involve a significant additional computational effort, and there is

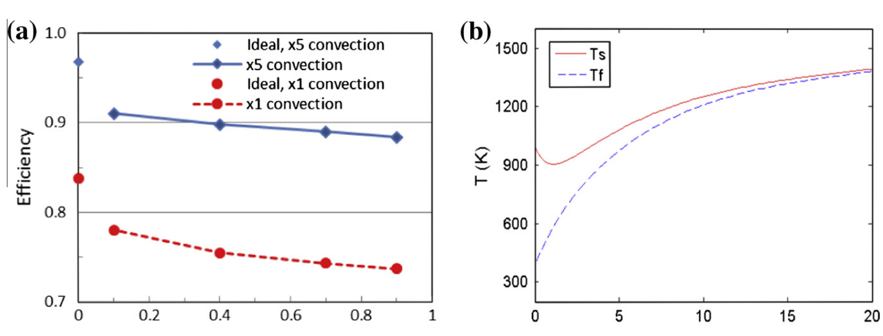

Fig. 9. (a) Efficiency as a function of the IR emittance; (b) temperature distribution for the case of an ideal selective absorber with 5× enhanced convection.

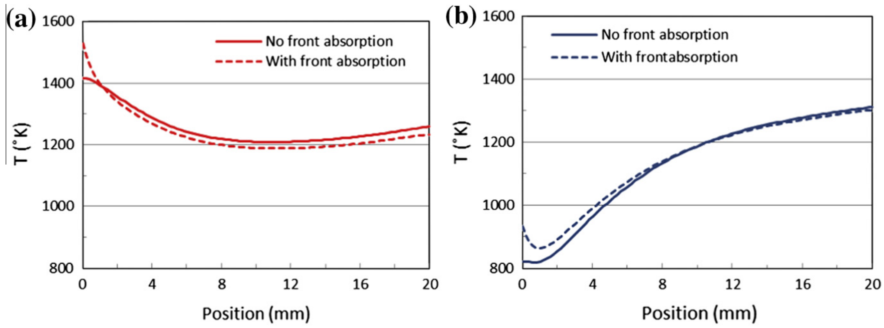

Fig. 10. (a) Solid absorber temperature distribution for the case $d = 4$ mm, $\phi = 0.9$, $k_s = 1$ W m^{-1} K^{-1} and no convection enhancement, (b) with 5× enhancement of convection.

no reason to neglect this effect. Note that the cases shown in Fig. 10 are of high porosity; in cases with lower porosity, more radiation will be absorbed at the front surface, and the local temperature rise at the front region will be higher.

3.7. The effect of back losses

For the cases shown above we applied an ideal boundary condition at the back wall, a perfect reflector and insulator that implies no loss of energy through the back wall. The general boundary condition in Section 2.3 allows estimating the loss associated with the back wall, if the wall properties are known. When considering real materials and geometry in a receiver where the hot air needs to be collected by an exit duct, some thermal losses are inevitable and should be taken into consideration. We estimated the magnitude of this loss by repeating the cases shown in Fig. 10 with the back wall defined as a gray surface with reflectance of 0.5 (representative of typical ceramic insulation), and temperature equal to the air outlet temperature (assuming high convective heat transfer between the exiting air and the back wall).

The loss due to the non-ideal back wall was in 3.4% for the case with no convection enhancement, and 0.8% in the case of 5× convection enhancement. The difference is due to the higher absorber temperatures that occur in the case of low convection, which lead to a higher radiation flux reaching the back wall, together with a lower air exit temperature. Losses of the order of 1% or higher are significant, and therefore the back loss should be taken into account in future analyses. The analysis should be based on a better definition of the exit duct and the representative temperature of the back wall, which has a significant impact on the magnitude of the back loss.

4. Discussion

The relatively simple model presented here can be used for fast simulation, parametric study, and optimization of the absorber performance. It models the major effects and physical mechanisms that influence the absorber performance, and therefore it can be expected to represent well the impact of changing design parameters. Using this model we have explored the absorber design parameter space and identified some interesting trends, which can serve as guidelines for future theoretical and experimental work. Clearly there is an additional range of phenomena that are created due to lateral variations in illumination and mass flux and by lateral boundary conditions; the one-dimensional model does not attempt to represent these effects, and they need a separate and more complex treatment.

A challenging aspect of the model is the use of effective medium properties, and in particular the volumetric convection coefficient. Existing knowledge and available correlations, based on both experimental data and detailed flow simulations, are inconsistent and contradictory as demonstrated in Appendix A. A further effort is needed to derive a consistent and reliable model of convective heat transfer through ceramic foams, which will address all relevant parameters (porosity, pore size, sample thickness), and provide valid results over a wide range of operational conditions. An effort in this direction was done (Kamiuto and Yee, 2005) but, as discussed in Appendix A, it is incomplete due to the neglect of the sample thickness length scale.

The effects of the boundary conditions, front and back surface, was considered in detail in contrast to previous studies that tended to idealize these conditions. We found that both boundary conditions have a significant effect, modifying the local temperature distribution near the surface, and contributing a loss that impacts the overall absorber performance. The energy balance at these boundaries should then be represented in detail.

The starting point for the study was the disappointing performance of current volumetric absorbers, which is much lower than the initial promise of the concept. The parametric study of absorber performance has revealed several insights that can be used to improve volumetric absorber designs. The absorber geometry has shown preference to higher porosity and larger pore size. The impact of both parameters is to reduce the rate of absorption of radiation, and spread the absorption process over a thicker layer of the absorber. This creates a larger volume where convective heat transfer takes place, and allows a gradual increase in temperature from the front towards the back of the absorber. However, geometric optimization is not sufficient to achieve high efficiency, and the best configuration shown here resulted in absorber efficiency around 76%. Manipulation of two additional properties is needed: a significant enhancement of convective heat transfer compared to the values that are naturally found in ceramic foams, and a reduction of thermal conductivity compared to the common materials currently used in absorbers. Assuming optimistic values for these two additional parameters, absorber efficiency of up to 88% was predicted, a significant improvement compared to the previous values, and close to the desired range of a high performance volumetric absorber. For the convection coefficient, improvement seems monotonic with no optimum point, although an asymptotic upper limit may exist. For the thermal conductivity there is an optimal value (about 1 W/m K for the case investigated here), created by the balance of the bulk volumetric effect vs. the effect of absorption at the front boundary.

Another major improvement in performance can theoretically be achieved with spectral selectivity. This is surprising since common wisdom claims that spectral selectivity should be effective only at moderate temperatures, and not for an absorber at temperatures around 1000 °C or much higher. The model showed that ideal spectral selectivity can bring the volumetric absorber efficiency very close to unity, and non-ideal spectral selectivity was predicted to provide up to 4% increase in efficiency relative to a gray absorber. Nevertheless, this effect is quite difficult to obtain with existing real materials. Further study and development is required in order to identify a material that can present this type of behavior.

A significant increase in convective heat transfer as proposed here is challenging, and will require a careful optimization of the pore-level structure of the absorber. This cannot be done with traditional methods of fabrication of ceramic foams, and will require innovative methods such as additive manufacturing. Some encouragement can be derived from the wide scatter of experimental data for convection coefficient: while the selected correlation in the current model was chosen to provide a conservative typical value, it seems that in some experiment the measured values were significantly higher, sometime by a factor of 2 or more. The conclusion is that some geometric structures provide strongly enhanced convection, due to differences that we cannot yet identify clearly, but should be amenable to careful future study. For example, for a pore diameter of 1 mm and $Re = 17$ (Fig. 11(a) of Appendix A), the volumetric Nu number is 1.73 according to the correlation provided by Fu, but 5.52 following the Kamiuto correlation. The media tested by Fu included YZA, SiC, Mullite and Cordierite, while the media considered by Kamiuto included Cordierite-Aluminite, Nickel, Alumina, Cordierite and Mullite.

The requirement of low thermal conductivity is needed in order to maintain the high temperature gradient implied by the volumetric effect, and avoid a high conductive heat flux towards the front of the absorber. This property can be achieved in principle using oxide ceramics, but these materials are usually not dark enough to serve as radiation absorbers. It is therefore necessary to develop new materials, combining low thermal conductivity, high absorptance of sunlight, as well as the other requirements usually applied to volumetric absorbers: resistance to high temperatures and thermal shock, mechanical stability, etc.

In all cases considered here the calculated pressure drop was less than 1 mbar, which is very small. Therefore fan work does not constitute a significant loss, and detailed pressure loss results were not reported here. This range of very low pressure drop is consistent with experimental evidence in volumetric absorbers. It should also be noted that following the conclusions of this work, preference should be given to larger pores, which leads to lower pressure drop across the absorber.

Variations in more than one dimension were intentionally omitted from the current model, in order to simplify and permit fast computation. However, in real solar concentrator systems the incident radiation flux, the temperatures, and the local air velocity are likely to have a non-uniform lateral distribution. In particular, the flow into the absorber will not be uniform as well, and in some cases this might lead to thermally induced flow instability (Kribus et al., 1996). The lateral non-uniformity of the temperature can increase the overall emission loss (Pitz-Paal et al., 1997). High thermal conductivity of the absorber mitigates this effect, in contrast to its detrimental effect on the volumetric temperature gradient into the absorber. An interesting possibility would be then to consider an absorber with anisotropic thermal conductivity in order to balance these contradicting effects. Additional effects not treated here include losses through sidewalls, and any asymmetric effects due to presence of flow inlets, etc. These effects must be analyzed using a more sophisticated model with two- or three-dimensional capabilities.

The study presented here was based on one typical case in terms of incident radiation flux, air flow rate, and air inlet temperature. Other cases with different boundary conditions will lead to a different range of absorber efficiency, but it is expected that the major trends identified here should also apply in these cases. This conjecture should be of course validated by repeating a similar study with different sets of boundary conditions. A more fundamental need for validation is the comparison of the model predictions vs. Experimental results, which is currently underway by the authors.

5. Conclusions

The overall conclusion from this parametric study is that volumetric absorbers that were constructed and tested to date have performed far below the desired level of efficiency, not due to some fundamental barrier that defines an upper limit on performance, but because their geometry and materials have not been properly optimized for maximum efficiency. In the past, commercially available ceramic foams have been used as volumetric absorbers, offering a limited range of features and materials that are not optimal for the application. It is then necessary to develop ceramic porous structures that are tailored to the application in terms of both geometric features and material properties. This will require a significant effort and poses major challenges in identification and synthesis of new materials, detailed optimization of absorber structure to achieve the ambitious performance goals, and development of new fabrications methods to achieve the desired new structures. Overcoming these challenges is essential in order to realize the promise of high-temperature, high-efficiency solar thermal plants. The analysis and parametric study, as presented here, can provide the guidelines and goals of such development, and help in achieving a significant improvement in volumetric absorber efficiency.

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Appendix A. Correlations for effective volumetric properties

Modeling of the porous absorber as a continuous homogeneous medium requires the definition of several averaged volumetric properties. Two of these properties pose a certain challenge since several conflicting correlations have been proposed in the literature to represent them. It should be noted that the averaged properties depend on the material properties and on details of the pore geometry, and it is often hard to generalize due to the wide range of foams, and the difficulty in foams testing. The purpose of this section is to review available models and to evaluate the sensitivity of the model predictions to the uncertainty in the foam averaged properties. The analysis refers to the volumetric heat transfer coefficient used in Eqs. (1) and (2), and the extinction coefficient used in Eq. (4).

A.1. Volumetric convection coefficient

The volumetric convection coefficient in ceramic foams was subject to many studies that produced different correlations, but did not agree even on the identification of the characteristic length scales to be used in the correlations. Various scales were used, including: a formal pore diameter based on manufacturer's specification, $d = 1/PPC$; a 'passage' diameter that includes the porosity, $d_m = \sqrt{4\phi/\pi}/PPC$ (Fu et al., 1998); and an effective strut diameter $D_s = d \cdot 2w/\sqrt{\pi}$ where $w = 0.5 + \cos\left[\frac{1}{3}\arccos(2\phi - 1) + 4\pi/3\right]$ (Kamiuto and Yee, 2005). Additional proposed length scales include an effective hydraulic diameter derived from the specific surface, and another derived from the permeability. In addition, a second length scale is the sample thickness, which some authors have ignored, but others have shown to be significant (Fu et al., 1998) Younis and Viskanta, 1993a. Some of the correlations include the porosity explicitly, others include it via its influence on the length scale, and some correlations are completely independent on porosity. In the following we will use the pore size d as the characteristic length for the definition of the dimensionless numbers Re and Nu_v, unless otherwise specified.

Experimental work on alumina foams has led to the following correlation (Younis and Viskanta, 1993b):

$$Nu_v = 0.137\left[1 + 33.73\left(\frac{d}{L}\right)\right]Re^{0.472\left[1+4.957\left(\frac{d}{L}\right)\right]} \tag{31}$$

This was based on measurements in the range $11.9 \leqslant Re \leqslant 798.1$ and $0.034 \leqslant d/L \leqslant 0.169$. The porosity was 0.875. The maximum deviation of the data from the fit was 14.3%. The same authors proposed a different correlation for a second type of alumina foam (Younis and Viskanta, 1993a):

$$Nu_v = 0.819\left[1 - 7.33\left(\frac{d}{L}\right)\right]Re^{0.38\left[1+15.5\left(\frac{d}{L}\right)\right]} \tag{32}$$

This was proposed for $5.1 \leqslant Re \leqslant 564$ and $0.005 \leqslant d/L \leqslant 0.136$, and porosity between 0.83 and 0.87. The maximum deviation was about 27.1%. Later work from the same group tried to generalize for several samples from different materials, with length scale d_m (Fu et al., 1998):

$$Nu_{v,d_m} = \left(0.0426 + 1.236\frac{d_m}{L}\right)Re_{d_m} \tag{33}$$

This applies in the range of $2 \leq Re_{d_m} \leq 836$ and $0.031 \leqslant d_m/L \leqslant 0.26$ for porosity between 0.742 and 0.916. The deviation was within ±20%.

A correlation that attempts to generalize a large amount of previous experimental results was proposed, based on strut diameter as length scale D_s (Kamiuto and Yee, 2005):

$$Nu_{v,D_s} = 0.124(Re_{D_s}Pr)^{0.791} \tag{34}$$

Range of applicability was $1 \leq Re_{D_s} \leq 1000$ and porosity between 0.742 and 0.948. The deviation was reported as ±40% for 78% of the data. However, the strut diameter, d_s was used as a single length scale, and the dependence on sample thickness that was reported in previous work is absent.

An alternative approach to developing convection correlations is to use flow simulations with detailed pore-level geometry. An advantage of the numerical approach is the ability to observe the spatial variation of the local heat transfer coefficient, in contrast to the previous empirical studies which only provided correlations for the average heat transfer coefficient across a full slab of porous medium. However, it is difficult to generalize the local Nu into an average value, as both the solid and fluid temperatures vary along the flow direction. Another limitation is that due to the large computational resources needed, only a small set of specific geometries is considered, and it is not known if similar results would be achieved with a different geometry. The geometry of a real RPC sample, measured by tomography, was used to derive the following correlation (Petrasch et al., 2007):

$$Nu_v = 1.5590 + 0.5954 Re^{0.5626} Pr^{0.4720} \tag{35}$$

This is valid in the range $0.2 < Re < 200$, $0.1 < Pr < 10$. However, only a single pore diameter (2.54 mm) and a single porosity (0.857) were investigated.

Another flow simulation was performed over an artificially designed periodic porous geometry (Wu et al., 2011), leading to a correlation was suggested for the fully developed region and is based on both Re and porosity, in the range $70 < Re < 800$ and $0.7 < \phi < 0.856$:

$$Nu_v = (32.504\phi^{0.38} - 109.94\phi^{1.38} + 166.65\phi^{2.38} - 86.98\phi^{3.38}) Re^{0.438} \tag{36}$$

Fig. 11 compares the correlations for three different pore diameters while the other parameters were fixed: $\phi = 0.9$, $L = 20$ mm, $Pr = 0.71$. The correlations were all converted to a consistent length scale of the formal pore diameter d. The correlations of Younis and Fu were obtained for foams thickness below 14 mm, and are extrapolated here for $L = 20$ mm. The gray vertical bar in each plot shows the range of interest in Re corresponding to the parametric study below.

We note that for each value of the pore diameter, the differences among the correlations are very large, in some cases more than a factor of 3. It is therefore necessary to select carefully which correlations seem to be more reliable than the others. First, two of the candidates are rejected based on the following arguments:

- Petrasch et al. (2007) – it is based on a single geometry with specific pore diameter and porosity, therefore its generalization is not supported by actual results; it uses a single length scale and ignores the sample length L; and for $d \geqslant 2.5$ mm, the trend in the Nu vs. Re is very different compared to all other correlations.
- Wu et al. (2011) – it is based on an artificial geometry that is not a real RPC; its stated range of validity is most cases outside the needed range of the parametric study; and in all cases, the correlation is overestimating the Nu compared to the others.

The remaining correlations of Younis and Viskanta (1993b), Fu et al. (1998) and Kamiuto and Yee (2005) seem to be in reasonable agreement for the larger pores $d = 2.5$ (differences are up to 52%), and $d = 4$ mm (differences of only 18%). However for the smaller pore size, the Kamiuto

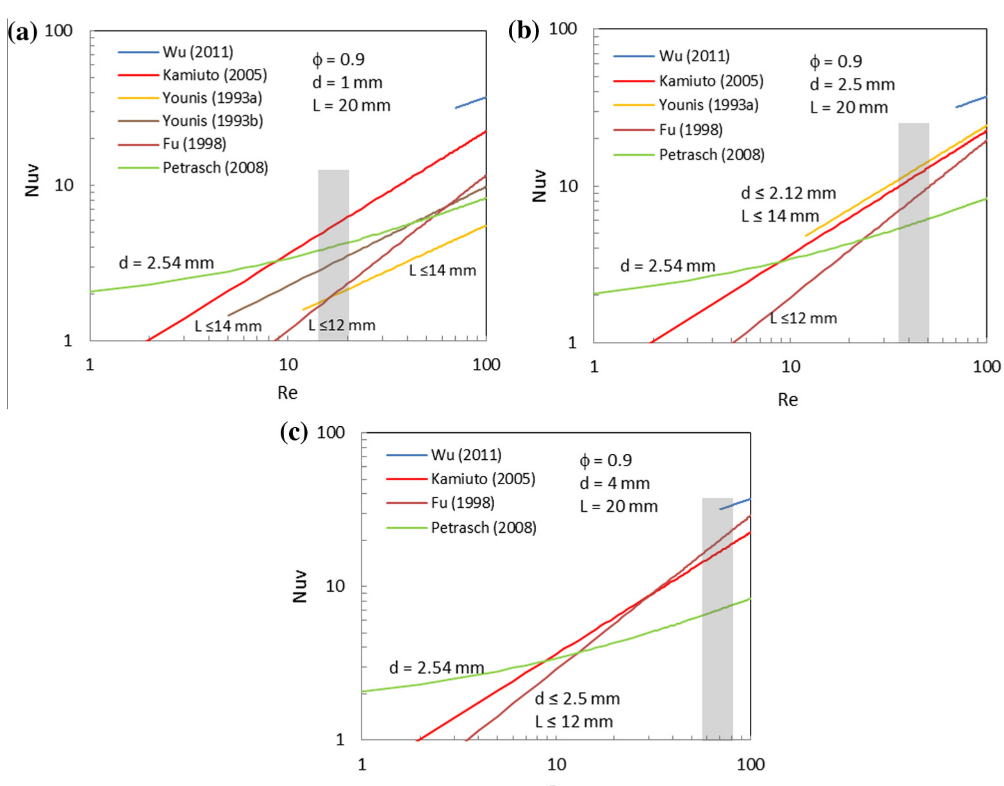

Fig. 11. Comparison of correlations for volumetric convection coefficient; gray bar is the Re range of interest for the parametric study (a) $d = 1$ mm, (b) $d = 2.5$ mm and (c) $d = 4$ mm.

correlation overestimates convection by a factor of about 2.5 within the range of interest compared to the other two correlations. A reason for this effect could be the absence of the second length scale in the correlation of Kamiuto. Therefore, we have selected the correlation of Fu, Eq. (3), as the one with the broader applicability, a wide base of experimental data, and better representation of relevant length scales. However, the correlation of Kamiuto should not be dismissed, since it is also based on a large amount of experimental data. We may conclude then that the selected correlation may be representative of convection in some typical ceramic foams, but in some other cases it may be possible to obtain much higher convection.

A.2. Extinction coefficient

The correlation used in the current study is given in Eq. (4), where we considered $C = 4.8$ based on empirical study on oxide-bonded (OB) SiC (Hendricks and Howell, 1996). The formulation of Eq. (4) is based on geometric optics analysis where the porous ceramic was considered as a suspension of mono-dispersed independently scattering spherical particles (Hsu and Howell, 1992). In this ideal case, the C becomes 3. Hendricks and Howell (1996) have suggested that Eq. (4) could be extended for actual foams simply by the adjusting the constant C. Thus, the approximation in Eq. (4) is that the C can be treated as a semi-empirical coefficient determined by the foam material and geometry.

The possible deviation between Eq. (4) and actual foams data is presented in Fig. 12: The semi-empirical coefficient C is shown by plotting the extinction coefficient β vs. the term $(1 - \phi)/d$. Eq. (4) is represented by the solid and dashed lines for $C = 4.8$ and $C = 4.4$, respectively. The data includes OB SiC and PS zirconia from Hendricks and Howell (1996), and alumina and SiSiC from Parthasarathy et al. (2012).

For the alumina, the C values lie between 3.3 and 5.3, with an average of 4.2 and a standard deviation of 0.7 (17%). For the OB SiC, the values lie between 3.6 and 5.3, with an average of 4.6 and standard deviation of 0.6 (12%). The best fit is at $C = 4.8$ (Hendricks and Howell, 1996). For the PS zirconia, the values lie between 4.75 and 6.75, with an average of 5.6 and standard deviation of 0.7 (13%). However, the best fit was considered at $C = 4.4$ (Hendricks and Howell, 1996). This level of uncertainty (12–17%) is reasonable considering the testing set-up and procedure limitations.

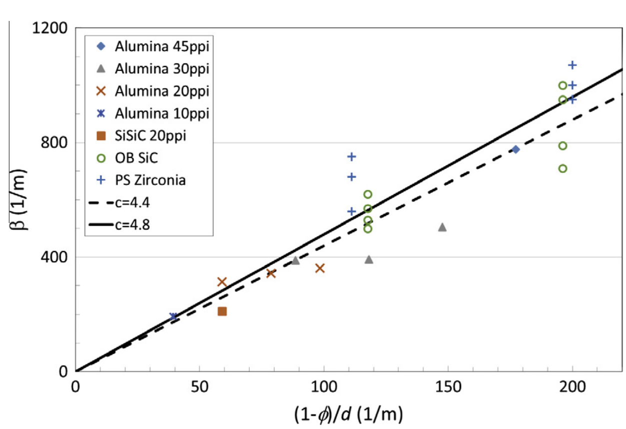

Fig. 12. Eq. (4) vs. empirical data.

Table 3
Sensitivity of absorber efficiency to variations in extinction empirical constant.

C	4.1	4.8	5.5
Absorber efficiency	0.771	0.764	0.756

The sensitivity of the absorber efficiency to changes in the value of the empirical constant in the extinction coefficient correlation was estimated by comparing the simulation of a representative case (OB SiC) with one standard deviation away from the proposed value, i.e., $C = 4.8 \pm 0.7$. The results are given in Table 3. The absorber efficiency shows a small decrease of 1.5% when the extinction coefficient increased by 34%, from 4.1 to 5.5. This decrease is expected since higher extinction coefficient means that more incident radiation is absorbed near the front of the absorber, and less is absorbed further into the depth of the absorber, leading to higher front temperatures and higher emission loss. However, it appears that the effect is small and the absorber has low sensitivity to fairly large variations in the empirical constant.

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